

The L^2 spectral geometry of the charge-one $SU(2)$ instanton moduli space on S^4

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Abstract

Let $\mathcal{M}_1(S_r^4)$ be the moduli space of charge-one anti-self-dual $SU(2)$ connections on the round 4-sphere of radius r , with its natural L^2 (Weil–Petersson) metric g_{L^2} . By Groisser–Parker and Doi–Matsumoto–Matumoto, $(\mathcal{M}_1(S_r^4), g_{L^2})$ is the interior of a compact smooth Riemannian 5-manifold with boundary: an $SO(5)$ -invariant, cohomogeneity-one metric on the closed 5-ball whose boundary is a totally geodesic round S^4 (the Uhlenbeck bubbling locus) at finite distance, of finite volume and strictly positive, non-constant sectional curvature. The same underlying 5-ball also carries the information (Fisher–Rao) metric, which is hyperbolic $\mathbb{H}^5$; the two metrics have different spectral type. We determine the L^2 spectral geometry. The Laplace–Beltrami operator is symmetric but not essentially self-adjoint on the open moduli space; its self-adjoint realizations are the elliptic boundary conditions on the bounding S^4 , and every such realization has purely discrete spectrum and obeys Weyl’s law, with leading constant computed explicitly. Separating by $SO(5)$ angular momentum reduces the problem to an explicit family of regular–singular Sturm–Liouville problems. A Liouville transformation, elementary here because $pw = g^4$, puts these in Schrödinger form with a potential increasing in both the radial variable and the angular momentum, and comparison against piecewise constant minorants of that potential yields certified two-sided enclosures. In the Groisser–Parker normalization the Friedrichs (Dirichlet) ground state is simple, $SO(5)$ -invariant, and lies in $[0.8203116, 0.8203177]r^{-2}$; the Neumann spectral gap lies in $[0.9988337, 0.9988386]r^{-2}$ and is the five-fold degenerate $k = 1$ mode, in the vector representation of $SO(5)$ (the instanton translation mode). We prove that this multiplet, and not any radial mode, is the gap. (The absolute values carry the metric normalization; the $SO(5)$ multiplet structure, the level ordering, and discreteness do not.) The boundary 4-volume is exactly $\frac{2}{3}\pi^6 r^4$. We also record the exact L^2 interaction of two separated bubbles, an ingredient for the higher-charge boundary geometry.

1 Introduction

The moduli space $\mathcal{M}_k(S_r^4)$ of charge- k ASD $SU(2)$ connections on S_r^4 carries a natural L^2 metric g_{L^2} , induced by the L^2 inner product on connection 1-forms. For $k = 1$ the underlying manifold is the open 5-ball, and the geometry of g_{L^2} was determined explicitly by Groisser–Parker [1] and, equivalently, by Doi–Matsumoto–Matumoto [2]; see also Franchetti–Schroers [3], who used it for the classical adiabatic dynamics of instantons on S^4 . We take that metric as given and determine the associated spectral theory of the L^2 -Laplacian, together with its organization under the isometry group $SO(5)$ and its interpretation as the collective-coordinate quantum mechanics of a single instanton.

The manifold underlying $\mathcal{M}_1(S_r^4)$ is the open 5-ball, as is hyperbolic space $\mathbb{H}^5$; the information (Fisher–Rao) metric on $\mathcal{M}_1$ is the hyperbolic metric [5]. The L^2 metric is a different $SO(5)$ -invariant

metric on the same ball: incomplete, of finite volume, and of strictly positive curvature, where $\mathbb{H}^5$ is complete and of constant negative curvature. The two Laplacians accordingly have different spectral type: that of $\mathbb{H}^5$ is purely continuous, $[4/r^2, \infty)$, while that of g_{L^2} is purely discrete. The L^2 spectrum is also the set of energy levels of the moduli-space (geodesic) approximation to a quantized single instanton, organized into $\mathrm{SO}(5)$ multiplets of dimensions 1, 5, 14, 30, 55, ...

Spectral theory for a Weil–Petersson metric on a moduli space is not new. Ji, Mazzeo, Müller and Vasy [4] treat the Weil–Petersson Laplacian on the Riemann moduli space of genus $\gamma > 1$, proving essential self-adjointness, discreteness of the spectrum, and a Weyl law. The present setting is the instanton analogue, and differs in one structural respect. There the metric is singular, with crossing cusp-edge singularities along a normal crossing divisor, and the Laplacian is essentially self-adjoint, so the self-adjoint realization is unique. Here $(\mathcal{M}_1(S_r^4), g_{L^2})$ completes to a *smooth* compact manifold with boundary; the Laplacian is correspondingly *not* essentially self-adjoint, and its realizations form the family of elliptic boundary conditions on the bubbling S^4 described in Theorem 2.1. The analysis is easier here for the same reason.

The metric. In $\mathrm{SO}(5)$ -invariant cohomogeneity-one coordinates, with $x \in (0, 1]$ a scale coordinate and the round S^4 the orbit of centers,

$$g_{L^2} = r^2 (f(x) dx^2 + g(x) g_{S^4}), \quad (1)$$

where g_{S^4} is the unit round metric and, in the normalization of [1, 2],

$$f(x) = \frac{4\pi^2}{(x^2 - 1)^4} \left(x^4 + 10x^2 + 1 - \frac{12x^2(x^2+1)}{x^2-1} \ln x \right), \quad (2)$$

$$g(x) = \frac{\pi^2}{2(x^2 - 1)^2} \left(x^4 - 8x^2 + 1 + \frac{24x^4}{x^4-1} \ln x \right). \quad (3)$$

The point $x = 1$ is a smooth interior point, the unique $\mathrm{SO}(5)$ -invariant instanton (the center of the ball), where the S^4 orbit collapses with $\sqrt{g} \sim (1 - x)$ and unit cap derivative (no cone angle). The limit $x \rightarrow 0$ is the bubbling boundary: a round S^4 of radius $\sqrt{g(0)} r = \sqrt{\pi^2/2} r$ at finite L^2 -distance, totally geodesic (the radius has vanishing normal derivative there). The metric is geodesically incomplete, with

$$\mathrm{Vol}(\mathcal{M}_1, g_{L^2}) \in [943.29, 943.50] r^5, \quad \mathrm{Vol}(\partial) = \frac{2}{3} \pi^6 r^4, \quad \mathrm{diam}_{\mathrm{rad}} = 3.7437 r, \quad (4)$$

and strictly positive, non-constant sectional curvature (isotropic value $\approx 0.2533 r^{-2}$ at the center, decreasing toward the boundary), in agreement with [1, Cor. C]. The volume enclosure is obtained from monotone Riemann sums for $\int \sqrt{f} g^2$, whose integrand is decreasing; it is not a recognizable closed form. The boundary 4-volume is exactly $\frac{2}{3} \pi^6 r^4$ because $g(0) = \pi^2/2$ exactly.

Remark 1.1 (On a hyperbolic misidentification). The hyperbolic metric on the same 5-ball is the information (Fisher–Rao) metric [5], *not* g_{L^2} . Being incomplete and positively curved, g_{L^2} has none of the continuous $[4/r^2, \infty)$ spectrum of $\mathbb{H}^5$; its spectrum is discrete (Theorem 2.1).

2 The spectral type of the L^2 -Laplacian

Let $-\Delta$ be the Laplace–Beltrami operator of g_{L^2} , initially on C_c^∞ of the open moduli space (functions supported away from the bubbling boundary).

Theorem 2.1 (Discrete spectral type and Weyl law). *The operator $-\Delta$ on $C_c^\infty(\mathcal{M}_1(S_r^4))$ is symmetric and bounded below but not essentially self-adjoint; its self-adjoint extensions correspond to elliptic boundary conditions on the totally geodesic bounding S^4 . Every such extension has compact resolvent, hence purely discrete spectrum $0 \leq \lambda_0 \leq \lambda_1 \leq \dots \rightarrow \infty$, with leading Weyl asymptotics*

$$N(\lambda) = \#\{n : \lambda_n \leq \lambda\} = \frac{\omega_5}{(2\pi)^5} \text{Vol}(\overline{\mathcal{M}}_1) \lambda^{5/2} (1 + o(1)),$$

where ω_d is the unit d -ball volume; in the normalization (1) at $r = 1$ the constant is $\frac{\omega_5}{(2\pi)^5} \text{Vol} = 0.50710$. There is no continuous spectrum.

Proof (sketch). The metric completion $\overline{\mathcal{M}}_1$ is a compact smooth manifold with boundary [1, Cor. C]; f, g in (2)–(3) extend smoothly with finite nonzero limits at $x = 0$ ($f(0) = 4\pi^2$, $g(0) = \pi^2/2$), and $x = 1$ is a smooth removable cap. Thus $-\Delta$ is uniformly elliptic with smooth coefficients on a compact manifold with boundary. Failure of essential self-adjointness on the interior C_c^∞ and the classification of extensions by boundary conditions on $\partial\overline{\mathcal{M}}_1 = S^4$ are the standard boundary theory. Compactness of the resolvent follows from the compact Sobolev embedding $H^1(\overline{\mathcal{M}}_1) \hookrightarrow L^2(\overline{\mathcal{M}}_1)$, giving discrete spectrum. The Weyl term is standard. $\square$

Remark 2.2 (On the second Weyl term). The refinement $\mp \frac{1}{4} \frac{\omega_4}{(2\pi)^4} \text{Vol}(\partial) \lambda^2$ (– Dirichlet, + Neumann), whose constant is 0.50734 in the normalization (1), holds under Ivrii’s condition that the periodic billiard trajectories carry measure zero. We do not verify that condition for this metric, and it is not automatic here: the boundary is a totally geodesic round S^4 , so the trajectories gliding along it are precisely the geodesics of a round sphere and are therefore all closed, though they form a measure zero set in the cosphere bundle. The interior flow is not analyzed.

3 SO(5) reduction and the low spectrum

Because g_{L^2} is SO(5)-invariant of cohomogeneity one, $L^2(\mathcal{M}_1)$ decomposes over the SO(5)-harmonics on the orbit S^4 . Writing an eigenfunction as $u(x) Y_k(\omega)$ with Y_k a degree- k spherical harmonic ($\Delta_{S^4} Y_k = -k(k+3) Y_k$, multiplicity $m_k = \frac{(2k+3)(k+1)(k+2)}{6} = 1, 5, 14, 30, 55, \dots$), the eigenvalue equation $-\Delta(u Y_k) = \lambda u Y_k$ becomes the regular-singular Sturm–Liouville problem

$$-(p u')' + k(k+3) \sqrt{f} g u = \lambda w u, \quad p = \frac{g^2}{\sqrt{f}}, \quad w = \sqrt{f} g^2, \quad x \in (0, 1), \quad (5)$$

with regularity at the center $x = 1$ (where $u \sim (1-x)^k$) and the chosen elliptic condition at the boundary $x = 0$. We solve (5) in three ways. Two are linear finite element discretizations that bracket the spectrum numerically: the conforming, consistent-mass scheme is Rayleigh–Ritz and bounds every eigenvalue from above, while the lumped-mass scheme converges from below, the two agreeing to width below 10^{-7} at 2.4×10^4 elements. The coefficients (2)–(3) are only C^1 at $x = 0$, carrying an $x^2 \ln x$ term that the Neumann modes feel and the Dirichlet modes, vanishing there, do not.

The third method is certified. Because $p w = g^4$, the Liouville transformation is elementary: with arc length $s = \int \sqrt{f} dx$ and $v = g u$, (5) becomes

$$-v'' + Q_k v = \lambda v \quad \text{on } (0, L), \quad Q_k = \frac{k(k+3)}{g} + \frac{g_{ss}}{g}, \quad L = 3.7437, \quad (6)$$

with v Dirichlet at the center, where $g \rightarrow 0$, and inheriting the chosen condition at the bubbling boundary, where $g_s = 0$ because that boundary is totally geodesic. Near the center $g \sim s^2$ and

$Q_k \sim 2/s^2$, the expected five-dimensional radial barrier. The function Q_k is increasing in s and increasing in k . Replacing it on each cell of a mesh by its value at the left endpoint therefore gives a piecewise constant $Q_k^- \leq Q_k$ exactly, and the resulting comparison problem, solvable in closed form cell by cell, bounds the eigenvalues from below by min–max. The deficit is first order in the mesh width, so the enclosures tighten linearly under refinement; those quoted below use 2×10^5 cells and have width below 10^{-5} .

Theorem 3.1 (Ground state, gap, and multiplet structure). *For the Friedrichs (Dirichlet) extension $-\Delta_D$ the ground state is simple, $\text{SO}(5)$ -invariant, realized by a positive eigenfunction, with (all absolute values in the normalization (1))*

$$\lambda_0(-\Delta_D) \in [0.8203116, 0.8203177] r^{-2} \quad (k = 0).$$

For the Neumann extension $-\Delta_N$ the ground state is 0 (the constant, $k = 0$), and the spectral gap is not an $\text{SO}(5)$ -invariant mode: it is the five-fold degenerate $k = 1$ mode

$$\lambda_1(-\Delta_N) \in [0.9988337, 0.9988386] r^{-2} \quad (k = 1, \text{ multiplicity } 5, \text{ vector representation of } \text{SO}(5)).$$

Both enclosures are certified. That the gap is this multiplet is likewise proved, not merely computed: Q_k in (6) increases with k , so the eigenvalues increase with k at each radial position, and the certified bounds

$$\lambda(k = 0, \text{ first overtone}) \geq 2.0351863 r^{-2}, \quad \lambda(k = 2, \text{ ground}) \geq 2.3973341 r^{-2}$$

separate the $k = 1$ multiplet from everything above it. The low eigenvalues λr^2 , as bracketed by the two finite element schemes, are:

k	m_k	Dirichlet (Friedrichs), lowest λr^2	Neumann, lowest λr^2
0	1	0.820318, 3.611916, 7.830235	0, 2.035539, 5.544165
1	5	2.180655, 5.667069, 10.584265	0.998839, 3.742777, 7.948320
2	14	3.938201, 8.125416, 13.743074	2.397606, 5.851182
3	30	6.096526, 10.986855	4.199088, 8.361619
4	55	8.657372	6.404425

We make no claim about closed forms for these eigenvalues. At the precision available here, about eight digits, inverse-symbolic searches return spurious matches against several bases, and are evidence in neither direction. What can be said concerns the equation rather than its eigenvalues: the coefficients (2)–(3) carry logarithms, so (5) is not hypergeometric.

Proof (sketch). Simplicity and positivity of the Dirichlet ground state, and its $\text{SO}(5)$ -invariance, are the Perron–Frobenius property of the Friedrichs Laplacian on a connected domain together with uniqueness of the positive ground state and invariance of the operator; a positive ground state is $\text{SO}(5)$ -fixed, hence $k = 0$. The Neumann ground state is the constant. The remaining statements are the numerical solution of (5) for $k = 0, \dots, 4$; ordering the union of the per- k spectra puts the five-dimensional $k = 1$ vector multiplet at $0.9988386 r^{-2}$, strictly below the first $k = 0$ overtone. Convergence data are in the ancillary computational files. $\square$

Remark 3.2 (Collective-coordinate quantum mechanics). In the moduli-space approximation [3], slow motion on $(\mathcal{M}_1, g_{L^2})$ is geodesic, and the associated quantum Hamiltonian is $-\frac{1}{2}\Delta_{L^2}$; the eigenvalues of Theorem 3.1 are its energy levels. One qualification is essential. Here $\mathcal{M}_1$ is the *unframed* moduli space, of dimension five (center and scale). The collective-coordinate quantization

standard in the soliton literature retains the $SU(2)$ framing, and it is the framing directions that carry the spin and isospin quantum numbers; those are absent here. The spectrum below is therefore that of the unframed reduction, not the full collective-coordinate spectrum. Within that reduction, the gap being the $k = 1$ vector multiplet says that the lowest excitation is a rigid translation of the center rather than a change of scale, the scale (breathing) excitation being the first $k = 0$ overtone, and the $SO(5)$ multiplet dimensions $1, 5, 14, 30, \dots$ are the degeneracies of the levels.

4 The two-bubble L^2 interaction (higher-charge boundary geometry)

For $k \geq 2$ the L^2 metric is that of a compact manifold with corners, the Uhlenbeck–Donaldson compactification adding finite-distance strata $\mathcal{M}_{k'}(S_r^4)$, $k' < k$; Theorem 2.1 generalizes verbatim (compact metric completion $\Rightarrow$ compact resolvent) to purely discrete spectrum with the $(8k - 3)/2$ -dimensional Weyl law. The content for $k \geq 2$ is the near-boundary geometry, governed by the L^2 interaction of separated bubbles, the instanton analog of the Gibbons–Manton asymptotic metric for monopoles. We record the leading interaction in closed form (as L^2 geometry, not spectrum).

Proposition 4.1 (Leading scale interaction at the maximally-bubbled corner). *Near the corner of $\overline{\mathcal{M}}_k(S_r^4)$ where k bubbles have small scales $s_1, \dots, s_k$ at distinct centers with pairwise chordal separations R_{ij} , the scale–scale block of g_{L^2} is, to leading order,*

$$(g_{L^2})_{s_i s_j} = 16\pi^2 \delta_{ij} + (1 - \delta_{ij}) \frac{48\pi^2 s_i s_j}{R_{ij}^2} + O((s/R)^4),$$

the diagonal the scale-independent BPST self-norm and the off-diagonal the two-bubble interaction of Lemma 4.2.

Lemma 4.2 (Exact two-bubble scale interaction). *Let A_1, A_2 be BPST instantons on $\mathbb{R}^4$ with scales s_1, s_2 , centers at separation R . Then*

$$\left\langle \frac{\partial A_1}{\partial s_1}, \frac{\partial A_2}{\partial s_2} \right\rangle_{L^2(\mathbb{R}^4)} = 48\pi^2 s_1 s_2 \int_0^1 t(1-t) \frac{2X(t) - t(1-t)R^2}{X(t)^2} dt, \quad X(t) = t(1-t)R^2 + (1-t)s_1^2 + ts_2^2,$$

which evaluates in elementary closed form with leading asymptotic $48\pi^2 s_1 s_2 / R^2$ and exact next-order coefficient -1 . (Machine-verified to ~ 2.5 ulp against the 4-dimensional integral.)

Remark 4.3. The closed form is correct as an L^2 integral; its earlier interpretation as a function of a hyperbolic distance referred to the information metric and is not used here.

References

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